COGNIZANT DIGITAL NURTURE 5.0 – DEEP SKILLING PHASE

WEEK 1 : Advanced SQL Mandatory Exercises

Submitted By

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Exercise 1: Ranking and Window Functions Goal:

Use ROW_NUMBER(), RANK(), DENSE_RANK(), OVER(), and PARTITION BY.

Scenario: Find the top 3 most expensive products in each category using different ranking functions. Steps: 1. Use ROW_NUMBER() to assign a unique rank within each category. 2. Use RANK() and DENSE_RANK() to compare how ties are handled. 3. Use PARTITION BY Category and ORDER BY Price DESC.

queries.sql

```sql
-- Create Table
CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50),
    Category VARCHAR(50),
    Price DECIMAL(10,2)
);

-- Insert Sample Data
INSERT INTO Products VALUES
(1, 'Laptop A', 'Electronics', 80000),
(2, 'Laptop B', 'Electronics', 75000),
(3, 'Laptop C', 'Electronics', 75000),
(4, 'Mobile A', 'Electronics', 50000),
(5, 'Chair A', 'Furniture', 10000),
(6, 'Chair B', 'Furniture', 9000),
(7, 'Table A', 'Furniture', 15000),
(8, 'Table B', 'Furniture', 12000);

```

```sql
-- Exercise 1 Solution
WITH RankedProducts AS
(
    SELECT
        ProductID,
        ProductName,
        Category,
        Price,

        ROW_NUMBER() OVER
        (
            PARTITION BY Category
            ORDER BY Price DESC
        ) AS RowNumber,

        RANK() OVER
        (
            PARTITION BY Category
            ORDER BY Price DESC
        ) AS RankValue,

        DENSE_RANK() OVER
        (
            PARTITION BY Category
            ORDER BY Price DESC
        ) AS DenseRankValue

    FROM Products
)

SELECT *
FROM RankedProducts
WHERE RowNumber <= 3
ORDER BY Category, RowNumber;
```

Output:

ProductID	ProductName	Category	Price	RowNumber	RankValue	DenseRankValue
1	Laptop A	Electronics	80000.00	1	1	1
2	Laptop B	Electronics	75000.00	2	2	2
3	Laptop C	Electronics	75000.00	3	2	2
7	Table A	Furniture	15000.00	1	1	1
8	Table B	Furniture	12000.00	2	2	2
5	Chair A	Furniture	10000.00	3	3	3

Exercise 2: Create a Stored Procedure Goal: Create a stored procedure to retrieve employee details by department.

Steps:

1. Define the stored procedure with a parameter for DepartmentID.

2. Write the SQL query to select employee details based on the DepartmentID.

3. Create a stored procedure named `sp_InsertEmployee` with the following code: CREATE PROCEDURE sp_InsertEmployee @FirstName VARCHAR(50), @LastName VARCHAR(50), @DepartmentID INT, @Salary DECIMAL(10,2), @JoinDate DATE

AS

BEGIN INSERT INTO Employees (FirstName, LastName, DepartmentID, Salary, JoinDate) VALUES (@FirstName, @LastName, @DepartmentID, @Salary, @JoinDate); END;

queries.sql

```sql
CREATE TABLE Employees (
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    DepartmentID INT,
    Salary DECIMAL(10,2),
    JoinDate DATE
);

INSERT INTO Employees
(FirstName, LastName, DepartmentID, Salary, JoinDate)
VALUES
('Nishant', 'Kumar', 1, 50000.00, '2024-01-15'),
('Rahul', 'Sharma', 2, 60000.00, '2023-06-10'),
('Priya', 'Patel', 1, 55000.00, '2024-03-20'),
('Amit', 'Singh', 3, 70000.00, '2022-08-05');

DELIMITER $$

CREATE PROCEDURE sp_GetEmployeesByDepartment(
    IN p_DepartmentID INT
)
BEGIN
    SELECT *
    FROM Employees
    WHERE DepartmentID = p_DepartmentID;
END $$

DELIMITER ;

CALL sp_GetEmployeesByDepartment(1);
```

Output:

EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
1	Nishant	Kumar	1	50000.00	2024-01-15
3	Priya	Patel	1	55000.00	2024-03-20

Exercise 5: Return Data from a Stored Procedure Goal:

Create a stored procedure that returns the total number of employees in a department.

Steps:

1. Define the stored procedure with a parameter for DepartmentID.

2. Write the SQL query to count the number of employees in the specified department.

3. Save the stored procedure by executing the Stored procedure content.

queries.sql

```sql
-- Create Table
CREATE TABLE Employees (
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    DepartmentID INT,
    Salary DECIMAL(10,2),
    JoinDate DATE
);

-- Insert Sample Data
INSERT INTO Employees
(FirstName, LastName, DepartmentID, Salary, JoinDate)
VALUES
('Nishant', 'Kumar', 1, 50000.00, '2024-01-15'),
('Rahul', 'Sharma', 2, 60000.00, '2023-06-10'),
('Priya', 'Patel', 1, 55000.00, '2024-03-20'),
('Amit', 'Singh', 3, 70000.00, '2022-08-05'),
('Rohit', 'Verma', 1, 65000.00, '2024-05-10');

```

```sql
-- Stored Procedure
DELIMITER $$

CREATE PROCEDURE sp_GetEmployeeCountByDepartment(
    IN p_DepartmentID INT
)
BEGIN
    SELECT
        DepartmentID,
        COUNT(*) AS TotalEmployees
    FROM Employees
    WHERE DepartmentID = p_DepartmentID
    GROUP BY DepartmentID;
END $$

DELIMITER ;

-- Execute Procedure
CALL sp_GetEmployeeCountByDepartment(1);
```

Output:

```
+--------------+----------------+
| DepartmentID | TotalEmployees |
+--------------+----------------+
|            1 |              3 |
+--------------+----------------+
```